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CAMBRIDGE INTERNATIONAL EXAMINATIONS

International General Certificate of Secondary Education

MARK SCHEME for the October/November 2013 series

0625 PHYSICS

0625/63 Paper 6 (Alternative to Practical), maximum raw mark 40

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1 **(a)** m = 180.2(0) and unit (g) [1]
V_1 value = m [1]
unit $\underline{cm^3}$ c.a.o. [1]

(b) V_2 = 170 c.a.o. [1]

(c) d_1 = 7.35 to 7.4, d_2 = 5.0 to 5.1, h = 7.9 [1]
D = 6.2 to 6.3 allow e.c.f. [1]
V_3 = 239 to 246 and 2 or 3 significant figures only allow e.c.f. [1]

(d) method 2 – one from:
some water left in cup/spilt
measuring cylinder not read at eye level/perpendicularly/bottom of meniscus
parallax explained [1]

method 3 – one from:
d_1 not at liquid level
d_1 and d_2 not inside diameters
difficult to measure h (because of sloping side)
h not measured at eye level/perpendicularly/parallax explained [1]

(e) mass of cup / zero reading on balance [1]

[Total: 10]

2 **(a)** **A** = 87(°C) and **B** = 88(°C) [1]

(b) units correct (symbols or words) [1]
times correct (0, 30, 60, 90, 120, 150, 180) [1]

(c) statement matching temperature changes (accept 'no significant difference' if justified) and justification matching statement (comparison of temperature changes) [1]
including specific mention of temperature change in same time [1]

(d) appropriate condition relating to comparison
i.e. any one from:
same size/thickness of beaker
same volume of water
same initial temperature
same room temperature / appropriate environmental condition
same time for cooling [1]

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(e) any sensible alteration e.g.
put lid on/cover top of **A**
extra experiment without insulation or lid / take lid off **B** [1]
matching explanation e.g.
most thermal energy loss by convection or o.w.t.t.e.
have only changed one factor or o.w.t.t.e. [1]

[Total: 8]

3 **(a)** correct symbol connected in parallel [1]

(b) **(i)** axes labelled, with units [1]
appropriate scales (plots occupying at least ½ grid) [1]
plots correct to ½ square [1]
best-fit line and thin, neat line, neat plots [1]

(ii) triangle method seen on graph [1]
large triangle (at least 1/2 candidate's line) [1]

(iii) *R* correct from *M* and in range 0.7 to 0.8 [1]
2 or 3 significant figures and unit Ω (symbol or word) [1]

[Total: 9]

4 **(a)** normal correct and pin separation at least 5 cm [1]

(b)(c) both reflected lines in correct place (through P_3, P_4 / P_5, P_6) and thin/neat [1]
θ = 40° within 1° [1]
θ = 62° within 1° [1]

(d) definite statement matching results (expect 'Yes' but allow e.c.f. if difference >10%)
and justification matching statement
(expect 'within the range of experimental accuracy' or o.w.t.t.e.) [1]
values from results shown/used (correctly w.r.t statement) [1]

(e) any two suitable precautions:
thin lines / fine pencil
view protractor perpendicularly/parallax explained
lines through centre of pin holes
pins well separated
pins vertical/not bent/viewed at base
place mirror so that reflecting surface is on line o.w.t.t.e. [2]

[Total: 8]

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5 **(a)** neat, clear table with column headings and correct units [1]
results arranged in order [1]

(b) **(i)** 40° [1]

(ii) plot a line graph [1]
reading will clearly not lie on line [1]
allow suggestion of appropriate mathematical treatment

[Total: 5]

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